KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

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FORMAT FOR STUDENT PROJECT PROPOSAL FOR THE 48th SERIES

OF STUDENT PROJECT PROGRAMME

(Handwritten proposals will not be accepted, please fill all the details in this MS word file, insert images / diagrams wherever necessary. Convert to pdf file, get it approved from the project guide / head of the department and principal of your institution. Keep ready the scanned pdf file of 1) Declaration and Endorsement 2) details of processing fees made and fill-up the Google Form.

https://forms.gle/ks2WxWB4ei1hgv9D9

1.	**Name of the College:** ST JOSEPH ENGINEERING COLLEGE.
2.	**Project Title:** DRY SOLAR PANEL CLEANING SYSTEM.
3.	**Branch:** ELECTRONICS AND COMMUNICATION ENGINEERING.
4.	**Theme:** NETWORKING AND COMMUNICATION ENGINEERING, EMBEDDED SYSTEMS, VLSI DESIGN, ROBOTICS
5.	**Name of project guide:** Name: Ms. Vijayalaxmi H Manjunatha Email id: vijayal@sjec.ac.in Contact No.: 9900232595
6.	**Name of Team Members:** Name: AKARSHAN V H USN No.: 4SO21EC014 Email id: 21i02.akarshan@sjec.ac.in Mobile No: 9986193121 

	Name: B. VIKAS **USN No.: 4SO21EC014** **Email id: 21h12.vikas@sjec.ac.in** **Mobile No.: 9901538923**  **Name: DEEVITH HEGDE** **USN No.: 4SO21EC019** **Email id: 21c20.deeving@sjec.ac.in** **Mobile No.: 8105448966**  **Name: MANISH** **USN No.: 4SO21EC056** **Email id: 21f27.manish@sjec.ac.in** **Mobile No.: 8792084161** 
7.	**Team Leader of the Project:** **Name: B. VIKAS** **USN No.: 4SO21EC014** **Email id: 21h12.vikas@sjec.ac.in** **Mobile No.: 9901538923**
8.	**Processing Fee Details :**
9.	**Date of commencement of the Project:** **23-9-2024**
10.	**Probable date of completion of the project:** **20-12-2024**

11. **Scope / Objectives of the project:**

- Remove dust and debris to enhance solar panel efficiency.
- Utilize a waterless, environment-friendly cleaning approach.
- IoT integration is used to automate the cleaning process remotely.
- Minimize maintenance costs and enhance system life.
- Ensure compatibility with various solar panel installations.

12. **Methodology:**

This system is able to automatically clean the solar panels. It uses a rotating brush and blower to move across the panel in order to remove dust and other debris. The movement is executed with precision and can traverse the entire surface. It is controlled by a mobile application, and users can operate it from anywhere. It works without water, so this is an eco-friendly, efficient way of cleaning and maintaining the solar panels, which ensures optimal performance.

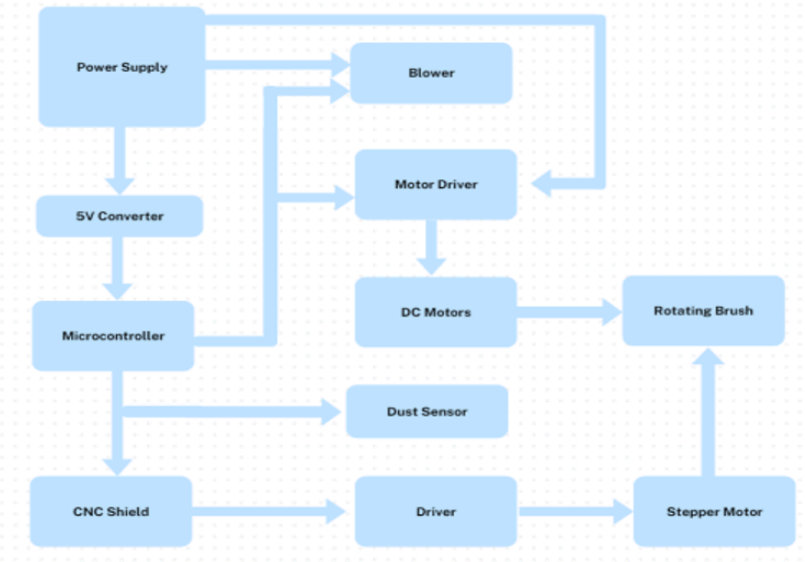


fig. 1 : Block diagram

- The system utilizes a CNC shield equipped with DRV8825 stepper motor drivers and stepper motors, thereby facilitating precise movement along the X and Y axes to guarantee comprehensive and efficient cleaning of the solar panel surface.
- The system, powered by ESP32 and Arduino Uno, integrates advanced automation so as to ensure synchronized control of movements of motors and the components that clean, ensuring efficient operation in the solar panel area.
- A rotating brush and blower are integrated for water-free cleaning, with a solution to remove dust and debris without losing the efficiency of the solar panel.

- The IoT-enabled system is operated by a smartphone application that allows real-time control, status monitoring, and user-friendliness to enhance the convenience and accessibility of cleaning solar panels.
- This innovative cleaning solution integrates advanced technology with mechanical precision, thereby establishing a sustainable and automated system that guarantees optimal maintenance of solar panels without necessitating water or manual intervention.

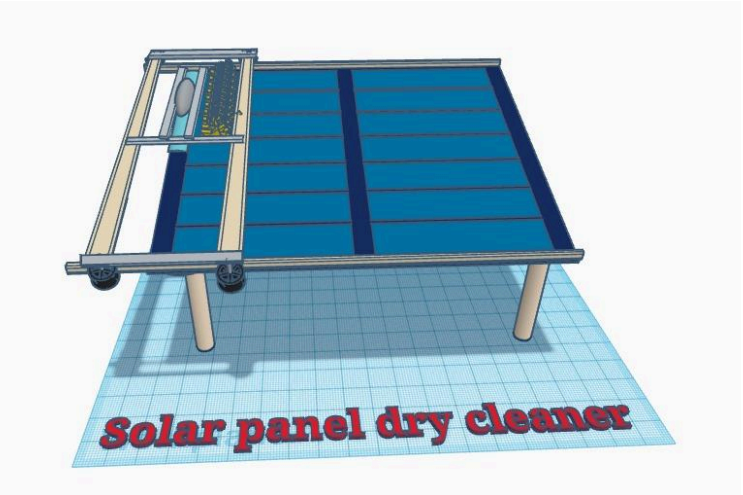


fig. 2: Prototype design

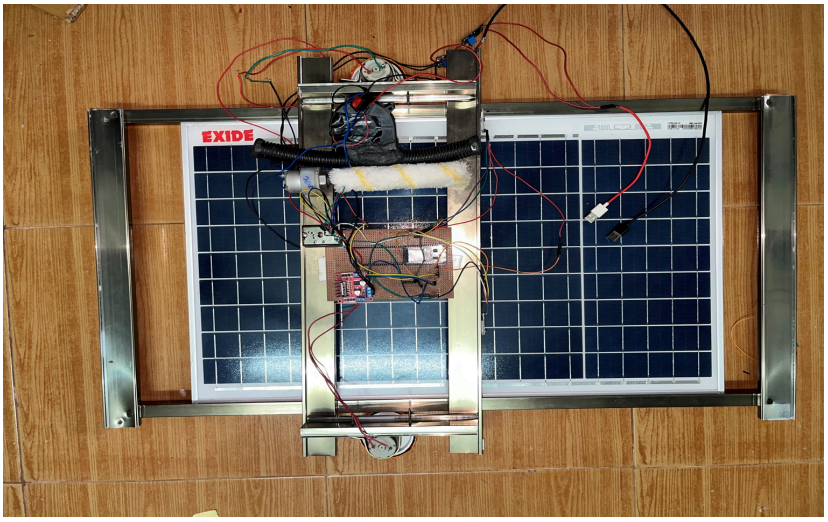


fig. 3: Project progress

The Automated CNC Solar Panel Cleaning System, So far we've been able to design and fabricate the aluminum frame and implement the X axis movement. This shows capability to deliver the project at hand. We will thus include a rotating brush equipped with a blower. In this proposal, completion and functional delivery of the Automated CNC Solar Panel Cleaning System is a target.

13.	**Expected Outcome of the project:** ● Improved Solar Efficiency, Enhanced power output by maintaining clean solar panel surfaces. ● Waterless Operation, the eco-friendly non-freshwater cleaning operation that saves water. ● Automated Functionality, IoT-based remote control and automation of scheduled cleaning ● Low labor and minimal maintenance inputs, hence low-cost operation. ● Scalable Design, It should be adapted for various solar installations and environment conditions.
14.	**Is the project proposed relevant to the Industry / Society or Institution?** **Yes / No:** Yes **If yes, please provide details of the industry / institution and contact details:** Yes, the dry solar panel cleaning system is highly beneficial to society, particularly in the renewable energy sector. However, dust accumulation on solar panels reduces their efficiency, leading to higher maintenance costs. The proposed automated dry cleaning system ensures the panels remain clean without water usage, addressing both efficiency and sustainability concerns. In addition to enhancing energy production and reducing manual labor, the system minimizes water wastage, which is especially critical in water-scarce regions. Furthermore, this technology has the potential to be adapted for use in space exploration, particularly in rovers deployed on other planets, where maintaining clean solar panels is essential for long-term missions.
15.	**Can the product or process developed in the project be taken up for filing a Patent?** **Yes / No:** No **Prior Art search done? Yes/No:** No

16.

Budget details:

Note: KSCST will provide nominal grant support for carrying out the project by students if selected by the project selection committee.

Budget	Description	Amount
a) Materials / Consumables	ESP-32	500.00
	Arduino UNO	350.00
	L298N Motor Driver	150.00
	MT3608 Booster Module	50.00
	Hall Effect Sensor Module A3144EUA	50.00
	24V Blower	800.00
	Batteries (rechargeable) for powering the system	1400.00
	12V Motor (4x)	1800.00
	CNC Shield A4988	300.00
	DRV8825 Motor Driver	300.00
	1.5A Stepper Motor (2x)	1800.00
	BH1750 Light Intensity Module	80.00
b) Labor	Student effort	250.00

	b) Travel	Transportation to purchase hardware	200.00
	c) Miscellaneous	Power adapters, Aluminium frame , screw , prototype casing, and unexpected minor expenses	1000.00
	Total		8130.00

17. **Any other technical details:**

- ESP32 is the primary microcontroller of the system. Programming has been done via the Arduino IDE. This is programmed for real-time monitoring and control.
- It uses a CNC kit for movement on the X and Y axes so that it sweeps every inch of the solar panel.
- Two Hall Effect sensors detect the magnetic fields at the panel edges, causing the motors to reverse their direction to prevent overrun and smooth movement.
- A rotating brush with 200 RPM motor fitted and blower so that dust is removed through it efficiently without the intervention of a human.
- The system has been designed to use minimal power; hence, it is suitable for large-scale deployment in solar farms or standalone installations.

18. **SPP Coordinator:**

Name: Mr. MANJUNATH B
Email id: manjunathb@sjec.ac.in
Contact No.: 9481818485

Name of the Project Guide:Ms. Vijayalaxmi H Manjunatha
Email id: vijayal@sjec.ac.in
Contact No.: 9900232595

Name of the HOD:Dr Dayakshini
Email id: hod.ece@sjec.ac.in
Contact No.: 0824-2868117

DECLARATION

We, the project team hereby declare that the details enclosed in the project proposal (Title of the Project: Dry Solar Panel Cleaning System , Branch: Electronics and Communication Engineering, College: St Joseph EngineerinG College) are true and correct to the best of our knowledge and belief. We undertake to inform the Karnataka State Council for Science and Technology (KSCST) of any changes to the project title or team members' names immediately through our project guide or the SPP Coordinator of our institution.

Additionally, we declare that the project work is original and not a result of copying or purchasing. We are committed to completing the project independently, with support from our faculty and project guide, while utilizing the facilities provided by the college. We will not engage in plagiarism and pledge to be sincere and dedicated in executing and completing the project as proposed.

We understand that if any of the above information is found to be false, untrue, or misleading, we may be held liable. We authorize the sharing of the project information contained in this proposal with KSCST, Bengaluru.

We acknowledge that the project team must exhibit and demonstrate the project, participate in the mid-term evaluation of sanctioned projects, and engage with experts. Additionally, we must exhibit the project at the Annual State-Level Poster Presentation and Exhibition, if selected. Should our team fail to participate in the mid-term evaluation or the Annual Exhibition (if selected), we understand that the supported project funds will be returned to KSCST.

We also enclose the endorsement form for KSCST, Bengaluru.

Name of the students with USN No.	**Signature with date**
1. Akarshan V H 4SO21EC007	
2. B Vikas 4SO21EC014	
3. Deevith Hegde 4SO21EC019	
4. Manish 4SO21EC056	

Ms. Vijayalaxmi H Manjunatha
Project guide
Email id: vijayal@sjec.ac.in
Contact No.: 9900232595

Dr Dayakshini
HOD - ECE
Email id: hod.ece@sjec.ac.in
Contact No.: 0824-2868117

ENDORSEMENT

This is to certify that 1) Mr. Akarshan V H , 2) Mr. B.Vikas , 3) Mr. Deevith Hegde , 4) Mr. Manish, are bonafide student(s) of Department of Electronics and Communication Engineering, in the degree program of our institution. If the project proposal submitted by these students under the 48th series of Student Project Programme is selected by KSCST, we will provide the requisite laboratory / Computer / infrastructure support in our college / Institution. Further we also take necessary steps to see that the project team will exhibit / demonstrate their project in the mid-term evaluation of project and in the Annual State-Level Poster Presentation and Exhibition (if selected). If the student team fails to send the completed project report or fails to attend the evaluation in mid-term evaluation of sanctioned projects or fails to attend the Annual State-Level Poster Presentation and Exhibition (if selected), the supported project amount will be returned to KSCST.

(Name & Signature of Project Guide with Seal)	**(Signature of HOD with Seal)**	**(Signature of the Principal with Seal)**
Email id: vijayal@sjec.ac.in	**Email id:** hod.ece@sjec.ac.in	**Email id:** principal@sjec.ac.in
Contact No.: 9900232595	**Contact No.:** 0824-2868117	**Contact No.:** 0824-2263732

DETAILS OF PROCESSING FEES MADE THROUGH NEFT / UPI PAYMENT

(**Note:** Include this page in the softcopy of the student project proposal. The student team shall furnish the details in the Google Form. It is informed to the students to 1) keep ready the softcopy of the project proposal and other documents and 2) Furnish the payment made details as processing fees and 3) update the details in the Google Form on the same day of payment made to KSCST by NEFT / UPI payment).

1. TITLE OF THE PROJECT	:	DRY SOLAR PANEL CLEANING SYSTEM
2. NAME OF THE TEAM LEADER	:	B VIKAS
3. EMAIL ID	:	21h12.vikas@sjec.ac.in
4. CONTACT MOBILE NO.	:	9901538923

PAYMENT MADE DETAILS

5. BANK REF. NO. / UTR NO. / UPI No. (12 digits)	:	433313096746
6. TRANSACTION ID	:	433313096746
7. NAME OF THE SENDER / ACCOUNT HOLDER and CONTACT NUMBER	:	B VIKAS 9901538923
8. NAME OF THE BANK	:	KARNATAKA BANK
9. PROCESSING FEES	:	Rs. 1,180/- (Inclusive of 18% GST)
10. DATE OF PAYMENT MADE	:	28/11/2024
11. TIME	:	10:32AM
12. MODE OF PAYMENT MADE (NEFT / UPI, PLEASE SPECIFY)	:	UPI (GPAY)

(Name & Signature of the team leader)

(Name & Signature of Project Guide or HOD with Seal)

KARNATAKA STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

Indian Institute of Science campus, Bengaluru

48th SERIES OF STUDENT PROJECT PROGRAMME (SPP)

(Note: This page is for information about bank details of KSCST to the student team and college / institution and not to include this page in the project proposal softcopy)

BANK ACCOUNT DETAILS OF KSCST

Name and address of the Institution	Karnataka State Council for Science and Technology, IISc Campus, Bangalore -560012
Account holder's name / Designation	Secretary, Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank, IISc Campus Branch, Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank, Indian Institute of Science, Bangalore-560012

BANK DETAILS

Name of the Agency	Karnataka State Council for Science and Technology IISc Campus, Bangalore - 560012
Account holder's name / Designation	Secretary , Karnataka State Council for Science and Technology
Bank Account No. & Name of the bank	Current A/C No. 0683201000024 Canara Bank IISc Campus Branch Bangalore-560012
IFSC Code	CNRB0000683
MICR Code	560015023
Bank Branch Address	Canara Bank Indian Institute of Science Bangalore-560012

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KARNATAKA STATE COUNCIL FOR SCIENCE

0683201000024@cnrb

BHIM BHARAT INTERFACE FOR MONEY | UPI UNIFIED PAYMENTS INTERFACE

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